

ADDER CIRCUIT

Condensed Study Notes

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Half Adders

When we add two individual bits together, there are four possible outcomes. A half adder is a circuit that can perform this addition.

Let's look at the possible combinations and their results:

- **A = 0, B = 0:** The sum (S) is 0, and the carry (C) is 0.
- **A = 0, B = 1:** The sum (S) is 1, and the carry (C) is 0.
- **A = 1, B = 0:** The sum (S) is 1, and the carry (C) is 0.
- **A = 1, B = 1:** The sum (S) is 0, and the carry (C) is 1.

Notice a pattern:

- The **sum bit (S)** is 1 only when the two input bits (A and B) are different. This is like a 'modulo-2 sum', where you add normally but then 'wrap around' to 0 if the result goes past 1. This operation is also known as an XOR (exclusive OR).
- The **carry bit (C)** is 1 only when both input bits (A and B) are 1. This is a straightforward AND operation.

So, a half adder uses an XOR gate for the sum and an AND gate for the carry.

COMPUTER ARCHITECTURE ADDER CIRCUIT

An adder is a fundamental digital circuit designed to perform addition.

It takes two bits, called **addends**, as input and produces an output representing their sum.

Adders can be connected together, or **concatenated**, to add binary numbers of any length.

There are two main types of adders:

- **Half Adder:** This circuit adds two single bits. As we've seen before, it produces a two-bit output: a **sum bit** and a **carry bit**.

Full Adder: *This circuit is more advanced. It adds two input bits plus* a **carry-in bit** that might be coming from a previous addition stage. Like the half adder, it also outputs a sum bit and a carry bit (which becomes the carry-out).*

Think of it like adding numbers by hand. When you add two columns of digits, you get a result for that column (the sum) and potentially a digit to carry over to the next column (the carry). A full adder handles this 'carry-over' idea within its circuitry.

Full Adders

A full adder is a circuit designed to overcome the limitations of half adders when dealing with multi-bit numbers. While a half adder can add two single bits and produce a sum and a carry-out, it cannot accept a carry-in from a previous addition.

A full adder solves this by adding three single bits together: the two numbers being added (like in a half adder) and a carry-in signal (Cin). This allows for the addition of multiple bits, as the carry-out from one full adder can become the carry-in for the next.

The output of a full adder is a two-bit result representing the arithmetic sum of these three one-bit inputs. This output consists of a sum bit and a final carry-out bit (Cout).

Full adders can be built using two half adders. You connect the two main inputs (A and B) to the first half adder. The sum output from this first half adder is then fed into the second half adder, along with the carry-in (Cin). Finally, the carry outputs from both half adders are combined using an OR gate to produce the final carry-out (Cout) for the full adder.

The logic expressions for a full adder can be derived and proven using truth tables, showing how the sum and carry-out bits are generated based on the three input bits.

Reference

This section serves as a reference point, providing links to external resources for further exploration of the topic of adders and related digital logic concepts. It is intended to guide you towards more in-depth information and supplementary materials.

For a deeper understanding of adders, including their implementation and applications, please refer to the following:

- [Wikibooks: Practical Electronics/Adders](https://en.wikibooks.org/wiki/Practical_Electronics/Adders)